C Programming Interview Questions & Answers - Part 1A (Q1-10)

1. What is the difference between malloc(), calloc(), and realloc()?

malloc() allocates memory without initialization. calloc() allocates memory for multiple elements and initializes it to zero. realloc() changes the size of previously allocated memory while preserving existing data whenever possible. All are declared in and allocated memory should be released using free().

2. What is a dangling pointer? How can it be avoided?

A dangling pointer points to memory that has already been freed. Accessing it causes undefined behavior and may crash the program. Avoid it by setting the pointer to NULL after free() and never using it after deallocation.

3. Difference between a wild pointer and a null pointer.

A wild pointer is uninitialized and contains a random address. A null pointer is intentionally set to NULL and points to no valid memory. Wild pointers are dangerous, while null pointers can be safely checked before use.

4. Difference between a structure and a union.

A structure allocates separate memory for each member. A union shares the same memory among all members. Structures allow simultaneous storage of all members, while unions save memory.

5. What is a memory leak?

A memory leak occurs when dynamically allocated memory is not released after use. It wastes memory and reduces performance. Always free allocated memory to prevent leaks.

6. Difference between call by value and call by reference.

Call by value passes a copy of the variable, so changes do not affect the original. Call by reference passes the address using pointers, allowing the function to modify the original variable.

7. What is a function pointer?

A function pointer stores the address of a function. It is useful for callbacks, menu-driven programs, and selecting functions dynamically at runtime.

8. Difference between stack and heap memory.

Stack memory is automatically managed and used for local variables. Heap memory is dynamically allocated using malloc()/calloc() and must be released with free(). Heap offers flexibility but requires manual management.

9. What is a segmentation fault?

A segmentation fault occurs when a program accesses invalid memory. Common causes include null pointers, wild pointers, array out-of-bounds access, and using freed memory.

10. What are storage classes in C?

Storage classes define the scope, lifetime, and visibility of variables. The main storage classes are auto, register, static, and extern, each serving different purposes.